AICHU TAN

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PROFESSIONAL SUMMARY

Data Scientist with M.S. in Big Data Analytics and experience developing machine learning and predictive modeling solutions using Python, SQL, and statistical analysis. Skilled in analyzing large datasets, building end-to-end ML pipelines, and delivering actionable insights through data visualization and dashboards. Experienced in collaborating with cross-functional teams to translate complex data into business and operational decisions.

TECHNICAL SKILLS

Programming: Python, SQL, R, JavaScript, C++
Machine Learning: XGBoost, Random Forest, Scikit-learn, Clustering, K-Prototypes
Deep Learning & Time Series: LSTM, ARIMA, Exponential Smoothing, TensorFlow, YOLOv8
Data & Visualization: Pandas, NumPy, Tableau, Matplotlib, ArcGIS
Cloud & MLOps: AWS (EC2, S3, RDS), Azure, GCP, MLflow, Git, SQL Databases

PROFESSIONAL EXPERIENCE

Data Science Researcher - SDSU DiMoLab — *Sep 2025 - Present*

- Developed **time-series forecasting models using Python** to predict outbreak trends, improving early-risk detection and enabling data-driven public health planning.
- Performed **data cleaning, feature engineering, and outbreak risk modeling** on epidemiological and climate datasets to improve predictive performance.
- Contributed to the development of an **AI-based early warning system for disease surveillance**
- Co-authoring a research manuscript on disease outbreak forecasting for submission to a peer-reviewed journal.

AI Researcher - Metabolism of Cities Living Lab — *Feb 2025 - Dec 2025*

- Designed an AI-powered plant disease detection and advisory system integrating YOLOv8 and LLM-based retrieval (RAG).
- Trained models on 13.5K+ images (20 classes), achieving 0.894 mAP@50.
- Developed a knowledge-driven recommendation pipeline using LangChain to convert agricultural research into actionable insights.
- Launched a real-time web application (Streamlit, Hugging Face Spaces) for smallholder farmers.
- Co-authored research accepted at the 4th International One Health Conference (Rome, 2025).

Business Owner - Wholesale Clothing — *Apr 2009 - Oct 2016*

- Increased sales by **50%** through data-driven market analysis.
- Optimized supply chain through forecasting and analytics.
- Managed import/export operations across Asia.

Process Engineer - ProMOS Technology — *Jul 2004 - Feb 2009*

- Improved DRAM yield via **statistical process control (SPC)** and root cause analysis.
- Reduced manufacturing defects through data modeling and process optimization.

SELECTED PROJECTS

End-to-End Dengue Forecasting System | *Python, LSTM, XGBoost, Random Forest, Streamlit*

- Built and deployed a time series forecasting system for weekly dengue cases using 15+ years of epidemiological, climate, and vector surveillance data.
- Engineered temporal and lag-based features and compared statistical and ML models to improve forecast accuracy.
- Identified key demand drivers (rainfall, humidity, lagged incidence) to support proactive planning decisions.
- Deployed an interactive forecasting dashboard for real-time exploration and monitoring.

Predicting Restaurant Ratings Using Yelp Metadata | *MLflow, XGBoost, RF, Pandas*

- Built a machine learning pipeline to classify restaurant success (≥4★ vs <4★) using 36,000+ Yelp business records.
- Engineered 61 features and applied **5-fold cross-validation**.
- Tracked experiments using **MLflow** to compare model performance.
- Optimized **XGBoost**, achieving **0.733 accuracy and 0.79 ROC-AUC**.
- Conducted feature importance and clustering to uncover consumer behavior patterns and restaurant archetypes.

Online Ticket Database Management System | *MySQL, AWS RDS, SQL, Python*

- Developed a full-stack ticket reservation system with cloud-hosted relational databases.
- Designed ERD schema and implemented normalized **MySQL** database on **AWS RDS**.
- Built a Python dashboard for ticket analytics and reporting.
- Automated SQL queries for reservations, seat availability, and revenue statistics.

Mapping for Change - Urban Inequality Analysis | *ArcGIS, Spatial Analysis, Data Visualization*

- Built an interactive ArcGIS dashboard to analyze **homelessness, housing affordability, and environmental inequality** across Downtown San Diego
- Integrated 5+ public datasets (ACS, 311, PITC, Tree Equity Score) and applied spatial analysis (heatmaps, choropleths) to identify inequality clusters
- Uncovered key relationships between **poverty, rent burden, and reduced tree equity**, highlighting areas of environmental injustice
- Delivered a public-facing web application to support **data-driven urban planning and policy decisions**

EDUCATION

M.S. in Big Data Analytics, San Diego State University - **GPA 4.0** — *Aug 2024 - Dec 2025*
Google Data Analytics Professional Certificate — *Dec 2022*
B.S. in Chemical Engineering, National Cheng Kung University (Taiwan) — *Sept 1999 - June 2003*